

Diagnosis and Assessment of Systemic Sclerosis

Disease Characteristics and Risk Factors

It occurs most likely in patients with the progressive diffuse form with a disease duration of < 4 years. The presence of **anti-RNA polymerase III antibodies** is considered a particular risk factor and is detected in about one-third of cases.

Diagnostic Criteria

In 2013, the American College of Rheumatology (ACR) and the European League Against Rheumatism (EULAR) published new classification criteria for systemic sclerosis. However, these criteria were primarily developed for clinical research purposes and cannot be applied to patients without skin involvement of the hands or to those with scleroderma-like disorders.

Capillaroscopy

Capillaroscopy (e.g., using a nail fold videocapillaroscope, stereomicroscope, or dermatoscope) is a well-established, non-invasive technique for identifying changes in the nail fold capillaries that help differentiate primary Raynaud's phenomenon from systemic sclerosis (SSc).

Clinical Skin Score

The **modified Rodnan skin score (mRSS)** is the best validated tool for assessing skin sclerosis progression. It is evaluated by manual palpation at 17 different anatomical sites. A score of 0, 1, 2, or 3 is assigned based on the degree of skin thickening:

- 0: Uninvolved skin
- 1: Mild thickening
- 2: Moderate thickening
- 3: Severe thickening

The total skin score is calculated as the sum of scores across all 17 areas. The mRSS is feasible, reliable, and validated for both initial and follow-up skin evaluation. Proper administration requires experience and adequate training.

Laboratory Parameters

Autoantibodies targeting specific nuclear antigens are a hallmark of SSc. The frequency of antinuclear antibodies (ANA) in SSc patients approaches 95%, consistent with reported rates of 85–99% in the literature. In one study, 86.6% of ANA-positive patients had SSc-specific antibodies, with 96.4% recognizing one of five major antigens: - Centromere

- Topoisomerase-1
- RNA polymerase III
- Exosomal ribonucleoprotein (PM/Scl)
- Uridine-rich small nuclear ribonucleoprotein (U1-snRNP)

These SSc-specific autoantibodies are generally mutually exclusive; co-occurrence in individual patients is rare.

Organ-Oriented Diagnostic Workup

A comprehensive baseline assessment—including clinical, laboratory, and radiographic evaluations—is essential to determine the patient's overall condition and organ involvement. Recommended diagnostic procedures are summarized in relevant guidelines. Lifelong follow-up is advised at least annually due to the chronic nature of the disease. In patients with progressive disease activity, more frequent monitoring is warranted.